

PHYS 267: Final Project

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Introduction

Since the launch of NASA's Kepler Space Telescope in 2009, scientists have gained access to high-precision records of brightness for thousands of stars. Originally launched to aid in exoplanet detection, data collected by Kepler has revealed far more than just that. The mission's photometry is optimized to detect shorter signals like planetary transits. This comes at the expense of long-term trends going unnoticed. However, as found by Montet and Simon in 2016, this information can be recovered through FFIs. FFIs or Full-Frame Images are sets of calibration data collected approximately once a month by the Kepler Space Telescope. Flux is recovered by drawing an aperture around the target star including all pixels that have a brightness that is greater than 1.5% of the peak brightness, extended with a 3-pixel border all around and adding the brightness of all the pixels in the aperture together (Montet et al., 2017). The dataset used for this analysis follows this procedure to extract flux across 53 FFIs collected through days 112.7 to 1557.5 of the Kepler mission.

KIC 8462852 is a main sequence F star 1480 light years away from the earth. The star is particularly well known due to its observed unusual behaviour in its brightness. Researchers found this particular star to undergo irregularly shaped, aperiodic dips in flux of up to 20%, a characteristic no other star exhibited. Nicknamed "WTF star" (Where's the flux?), "Tabby's star" or "Boyajian's star", the star gained rapid public traction after the release of Boyajian et al.'s paper about it titled "Planet Hunters X. KIC 8462852 – Where's the flux?" in 2016.

Separately from these short-term dips, Montet and Simon's 2016 paper conducted FFI-based photometry on the star and found the star to dim by 0.9% over the first 1000 days and a much more rapid dip was observed in the following 200 days (2%). The study compared this result to a sample of 193 nearby comparison stars and 355 stars with similar stellar parameters, of which none were found to exhibit such a rapid dip in their brightness.

The goal of this project is to analyze a dataset consisting of relative flux measurements for Boyajian's star to determine whether the star's flux shows a significant increase or decrease. In this analysis, time is treated as an independent variable and flux as a dependent variable. Based on the 2016 paper by Montet and Simon, we hypothesize that the star's observed flux exhibits a statistically

significant decline over time. This will be tested against the null hypothesis that the observed flux does not change significantly with time.

Results

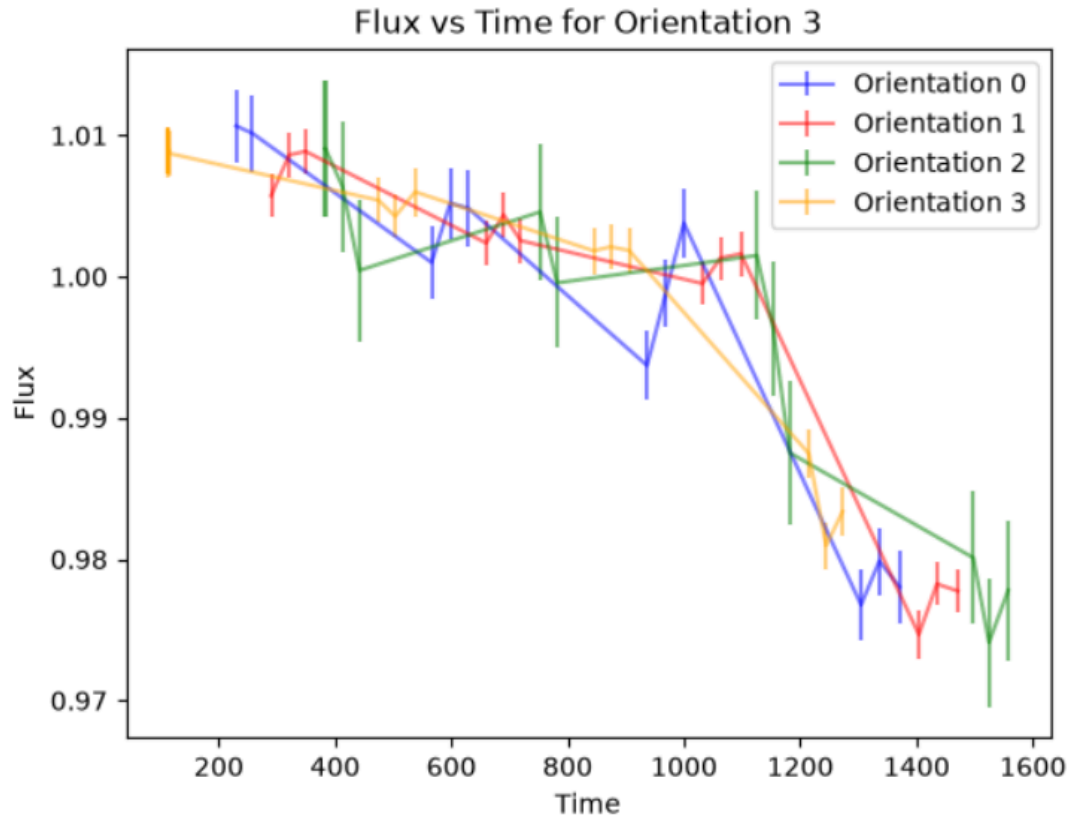


Figure 1: A coloured line graph representing the measured flux over time with each point containing errorbars, separated by each orientation labeled and coloured separately.

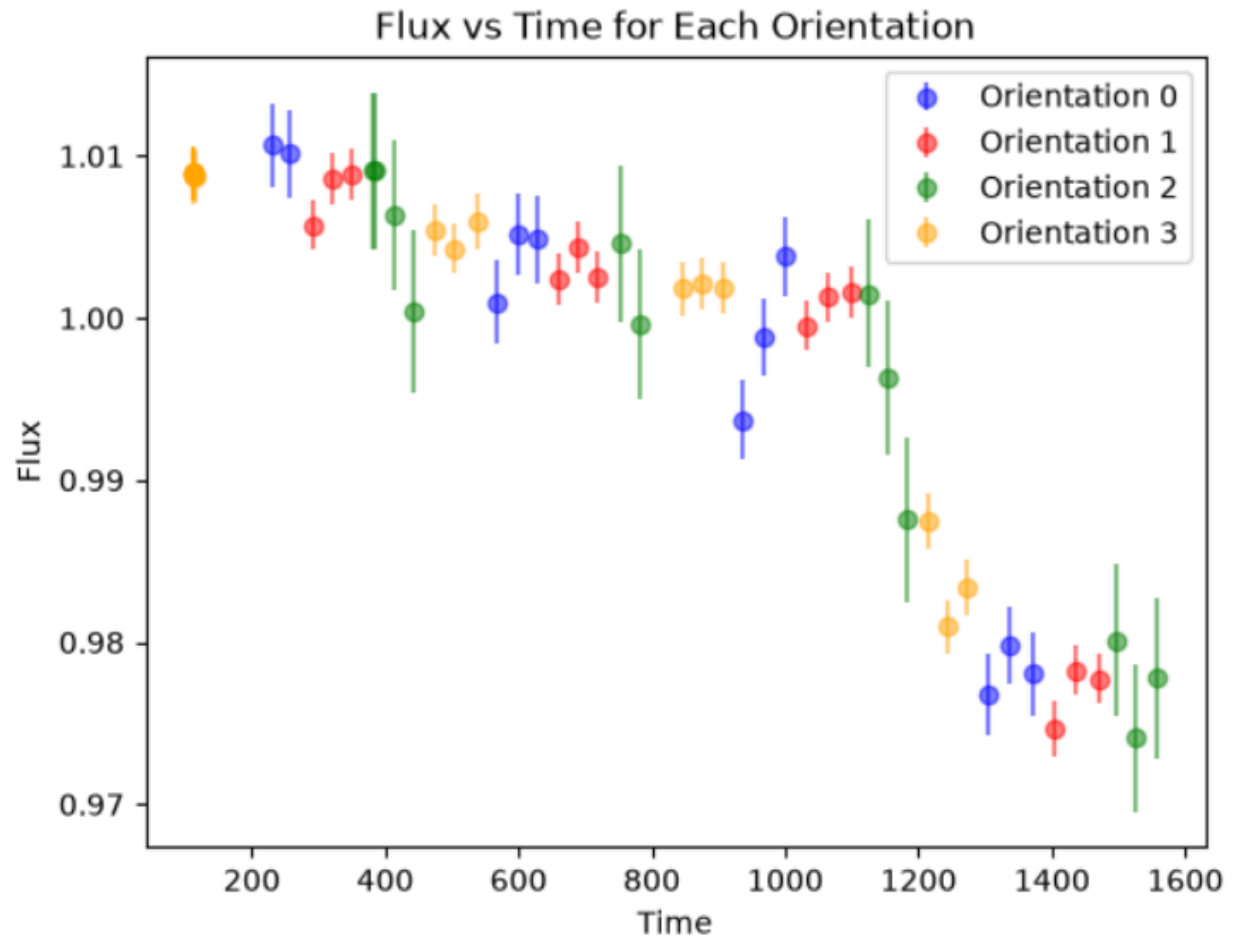


Figure 2: A scatter plot representing the measured flux over time with errorbars for each orientation, with each separately coloured with orientation 0 being blue, 1 being red and so on.

Flux Measurement Share by Orientation

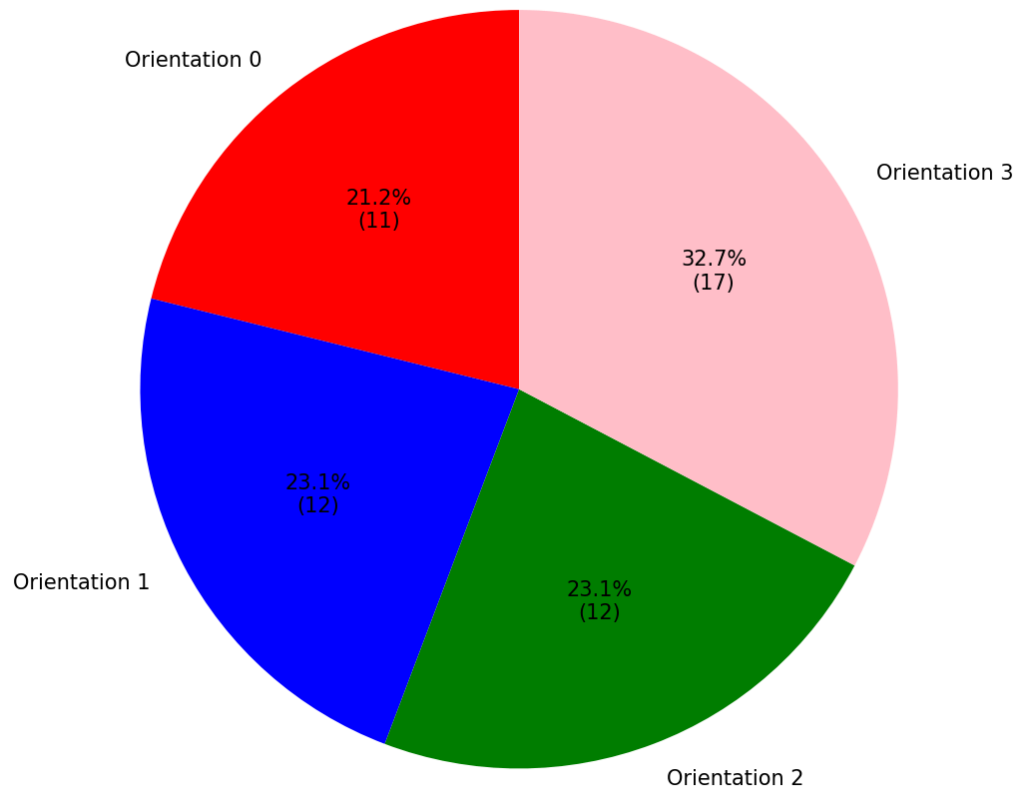


Figure 3: Pie chart representing the number of measurements taken for each orientation with respective percentages.

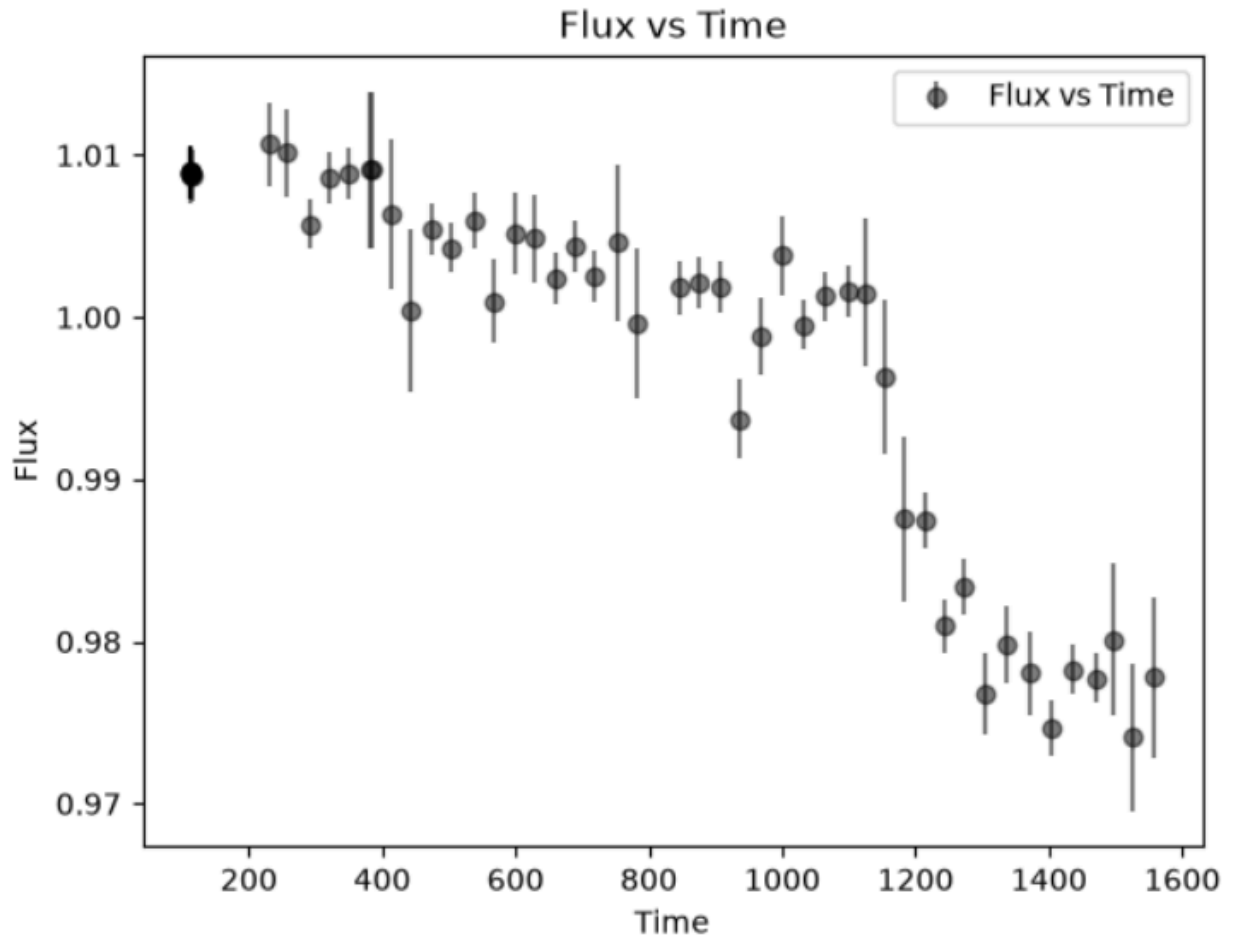


Figure 4: A scatter plot representing the measured flux over time with errorbars with all the orientations combined.

The best method to represent this data would be in the form of a scatter plot with the respective uncertainty measurements and differentiation in the orientations, such as in figure 2. This is because it gives a more cohesive and interpretable visualization of the relationship between the flux with respect to time for each of the orientations without covering the data with multiple lines connecting each point. It has the benefit of having each orientation labeled and coloured separately from the others whilst also maintaining its legibility.

Analyses

Qualitative Representation of Data

Figure 2 plots relative flux as a function of time for KIC 8462852, with measurements separated into four subsets corresponding to the four distinct spacecraft roll orientations that Kepler cycled through across its observing quarters. Each point represents a flux measurement with an associated uncertainty, and flux is normalized such that a value of 1.00 corresponds to the star's nominal, undimmed brightness. Time spans roughly 100 to 1,550 units.

In figure 2, there is a visually significant decrease in the flux across all orientations over the time, this would go in line with our predictions according to Montet and Simon's paper in 2016. In the earlier times the flux values cluster together which would indicate that the star is close to its nominal brightness near the beginning of the measurements. Nearing the end of the observations the flux dims by approximately 3% compared to its flat baseline prior. The fact that this overall trend is apparent in all 4 orientations suggests this behavior is astrophysical and not an issue with the technology used.

The error bars associated with each point are generally small relative to the size of the overall trend, particularly for Orientations 0, 1, and 3, whose points fall along a comparatively tight, coherent trajectory. Orientation 2 shows somewhat larger uncertainties and more scatter at several epochs, but its overall trajectory still tracks the same downward trend seen in the other three orientations, and its points overlap with the others within uncertainty at most shared time intervals. There is no evidence of a systematic offset between orientations that would suggest the dimming is an artifact of how the star fell on the detector in a particular roll configuration; rather, the four subsets appear to be sampling the same underlying astrophysical signal, just with somewhat different noise characteristics.

Qualitatively, what can be visually extrapolated from the data in figure 2 is in agreeance with the hypothesis by Montet and Simon (2016), which argues against the data being measured noise. However with that being said, the visual aspect of measurement cannot solely prove the decrease in flux over time is scientifically significant, however this does heavily suggest that the drop in flux is significant.

Quantitative Representation of Data

A summary of our data and distributions will be done by calculating the sample size, mean, median, standard deviation, variance, minimum, maximum, and the range.

In the context of this dataset, the mean represents the average brightness of the star during the observation period. Comparing the mean and the median will tell us how symmetric the distribution is. If they are similar, the distribution is approximately symmetric. If there is a big difference, the distribution could be skewed by large increases or decreases in the stars' brightness.

For our data, we got values of 0.998446 and 1.002245 for the mean and the median, respectively. Comparing these two values, we can see that they are quite similar, this suggests that the distribution is approximately symmetric. However, after plotting the data, it is clear that there is no symmetry in this dataset due to the larger dips in the measurements.

The standard deviation will tell us how much the brightness changes during the observation period. The variance measures how spread out the flux values are from the mean. A small variance would tell us that the flux remains consistent during the observation period, while a larger variance tells us there is a lot of variability in the brightness of the star.

The standard deviation and variance were measured to be 0.011660 and 0.000136, respectively. These values tell us that the flux measurements are clustered around the mean value and that the star's brightness was fairly consistent during the observation period, with some small variability.

The maximum and minimum values will provide us with the range and show the total spread and what the largest and lowest brightness is.

The maximum and minimum flux values are 1.01067 and 0.97411, respectively, and give us a range of 0.03656. When looking at the plot, we can see that the flux drops more from the mean than it increases. This shows that there were periods where the brightness was noticeably lower and explains the variability in the brightness.

Overall, these measurements show that the flux remained relatively stable throughout the observation period, with small variability.

Model and Fitting Data

The provided dataset was fitted with 3 different models using python. First a linear fit was applied to the dataset. Montet and Simon's 2016 paper found the linear fit for the first 1000 days to have a slope of -0.341 with an uncertainty of 0.041%. Our linear model fitted with a slope of -0.8189 and an intercept of 1.01556, or +1.556% relative to the normalized baseline of 1.0. Although this slope is steeper than the value provided by Montet and Simon, this is expected as we are instead modelling the whole dataset, which includes the steeper decline observed in the next 200 days. Checking the goodness-of-fit with a chi-squared test we found the model to correspond to a value of 365.408. For a good fit, we expect a value of chi-squared close to the degrees of freedom (50). We also found this model to correspond to a coefficient of determination of $R^2=0.7993$, meaning it accounts for about 80% of the variance in the normalized flux, however, considering both chi-squared and R-squared together, we can see that this model is still unfit to effectively describe our data.

We continued modelling this data, this time instead fitting a piecewise linear model. Montet and Simon's paper mentioned 3 different trends: the first 1000 days observed a small decline, followed by a steeper decline in the next 200 days, followed by a roughly constant flux in the final 200 days. Although, the paper also notes that the data from the final 200 days is still consistent with the first 2.7 years. Following this, a 3-segment piecewise linear plot was fitted to the data. Montet and Simon's paper overlapped their 3 segments; instead, a simpler approach was used here with no overlap between segments. The first segment focused on days 112 to 1097 and found a slope of -0.3767 ± 0.042 %/year, consistent with Montet and Simon's value of -0.341 ± 0.041 %/year. The second segment focused on the next 150 days of data and found a slope of -6.047 ± 0.9358 %/year, this is almost double the reported value of $3.37 \pm 1.48\%$ yr⁻¹ from Montet and Simon's paper. The driving cause of this discrepancy is most likely the difference in analysis tools. Using a much simpler model, the fit varies largely due to the lack of datapoints (5 total) within the chosen range. The last segment focused on days 1250 to the end of the dataset and found a slope of -0.452 ± 0.3218 %/year, this is consistent with both of the paper's claims, this is a relatively low slope, which corresponds to minimal changes in flux over the short time period of the third segment and is consistent with the first 2.7 years of data. Overall, this model corresponds to a chi-squared value of 40.97. Resulting in a reduced chi-squared value of 0.891, this shows that

the piecewise linear model fits the data better than a simple linear model. This is further backed-up by an R^2 of 0.9649.

Finally, a quadratic model was fitted to the data. The quadratic model ($f(t) = at^2 + bt + c$) found a negative leading coefficient of $-2.031e-08 \pm 2.947e-09$, corresponding to a concave-down parabola with a b-value of $9.426e-06 \pm 4.761e-06$ and a c-value of 1.00739 ± 0.00156 . Overall, this model corresponds to a chi-squared value of 150.837 (much higher than the 49 dof) and an R^2 of 0.8981 (a 10 percentage point increase from the linear model but still lower than the piecewise linear model). Therefore, although the quadratic model fits the data better than the linear model, the piecewise linear model is still the best fit found for our data.

Hypothesis Test #1: Does the normalized flux change significantly over time?

Hypothesis: There is a significant linear relationship between the observation time and normalized flux. The regression line slope is not zero.

Null Hypothesis: There is no significant linear relationship between the observation time and normalized flux. The regression line slope is zero.

We will use a linear regression test with time as the independent variable and flux as the dependent variable to see if there is a significant relationship between observation time and normalized flux.

The significance of the regression will be evaluated using a two-tailed t-test, with a significance level of 0.05.

After doing a linear regression and collecting a p-value of $4.56 * 10^{-19}$, we can reject the null hypothesis and conclude that the slope is not zero since the p-value is smaller than our significance value. The collected slope value is $-2.242 * 10^{-5}$, with a standard error of $1.589 * 10^{-6}$. Since the slope is negative, the flux is decreasing over time.

The regression produced a t-value of -14.11, this shows that the measured slope value is 14 standard errors below zero, and since it is a large value, this provides more evidence that we can reject our null hypothesis. Additionally, a R^2 value of 0.7993 was produced from the linear regression. This shows that approximately 80% of the data can be explained by a linear relationship.

Hypothesis Test #2: Comparing flux across different orientations

Hypothesis: Mean flux differs significantly between at least one pair of orientation groups

Null Hypothesis: Mean flux is the same across all four orientation groups

To start, the data was split into 4 different groups according to orientation (0, 1, 2, 3). Next, each group was tested for normality using `scipy.stats.normaltest()`. We found all of the orientation groups to pass the test ($p > 0.05$) except orientation 3 with a p-value of 0.0115:

Orientation	Stat	P-Value	Normal? ($p > 0.05$)
0	2.1525	0.3409	Yes
1	3.2420	0.1977	Yes
2	2.1071	0.3487	Yes
3	8.9318	0.0115	No

Therefore, to be safe and cross-check, both the One-Way ANOVA test and the Kruskal-Wallis test were used with $\alpha=0.05$ to compare all the groups.

The One-Way ANOVA test was applied to the 4 groups using `stats.f_oneway()`, resulting in an f-statistic of 1.1160 and a p-value of 0.351900. Since our resulting p-value is greater than 0.05, we fail to reject the null hypothesis. Therefore, there is no significant difference in the mean across all 4 orientation groups

Next, the Kruskal-Wallis test was used to cross-check and confirm that our result is not dependent on the assumption of our data following a normal distribution. Using `stats.kruskal()`, we find an H-Statistic of 4.8811 and a p-value of 0.180717. Once again, since the p-value is greater than 0.05, we fail to reject the null hypothesis.

Therefore, since both tests lead to the same conclusion, we can be confident in concluding that there is no statistically significant difference in the mean flux across the four orientation groups and that our conclusion is not dependent on whether our data follows a normal distribution. This is a meaningful conclusion as we can now be certain that the existence of the unusual dimming events in KIC

8462852 are not a result of the Kepler Space Telescope. This can also help us gain further confidence in our models from analysis 3.

Discussion

Qualitative

By viewing the figure provided in the results section (Fig. 2) by qualitative analysis, the approximate 3% drop in flux appears to be in agreeance with the hypothesis stated in the introduction along with the claim made by Montet and Simon (2016). This significant drop in the flux over time is shown across all four orientations with all uncertainty values between said orientations being relatively small for the size of the overall trend, more specifically in orientations 0,1, and 3. And while orientation 2 has a larger uncertainty, it still follows the same general trend which suggests that the hypothesis given the data was correct.

Quantitative

The quantitative analysis shows that there was some variability in the normalized flux during the observation period. The mean flux was measured to be 0.998446, and the median was 1.002245. While these two values are quite similar, the data is not symmetrical due to larger dips in the measurements. The low standard deviation and variance values, 0.011660 and 0.000136, respectively, show that most measurements were relatively close to the mean with very small variability. The minimum and maximum flux values were 0.97411 and 1.01067 and give a range of 0.03656. The large drops below the mean suggest that the star experienced more periods where the brightness was noticeably lower.

Modelling

Beyond describing the overall distribution, it is also important to assess how well the data can be modelled using known mathematical functions. Out of the three tested models, the 3-segment piecewise linear model provided the best overall fit with the lowest χ^2/dof value and the highest R^2 value. This suggests that the flux of Boyajian's star transitions between "phases" where its flux's decline rate changes between phases. This model is consistent with the one reported by Montet and Simon (2016). The slope of the first segment was found to be -0.3767 ± 0.042 %/year, consistent with Montet and Simon's finding of -0.341 ± 0.041 %/year within 0.61σ . The second segment's slope of -6.047 ± 0.9358 %/year is almost double the paper's reporting of $3.37 \pm 1.48\%$ yr⁻¹. However, both remain statistically consistent within 1.53σ and this can mostly be attributed to different statistical tools being used over a small set of data consisting of just 5 points. The final segment models the remaining set of data with a slope of -0.452 ± 0.3218 %/year, consistent with

Montet and Simon's claim of this being consistent with the first 2.7 years. Overall, the piecewise linear model's strong fit suggests the existence of phases in relation to the star's flux and its dimming.

HypoTest 1

The linear regression showed strong evidence of a linear trend over time. The slope was calculated to be $-2.242 * 10^{-5}$, showing that normalized flux decreased over time. The t-value was -14.11, which shows that the slope is approximately 14 standard errors below zero. The p-value was $4.56 * 10^{-19}$, which is much smaller than our significance level of 0.05, so we can reject our null hypothesis and conclude that there is a significant linear relationship between time and normalized flux.

HypoTest 2

Finally, a sanity check was conducted to conclude if the measured flux pattern was a result of the orientation of the Kepler Space Telescope. First, the dataset was divided into 4 different groups according to the orientation (0, 1, 2 and 3). After checking all the groups for normality, both the One-Way ANOVA and the Kruskal-Wallis Test were applied. Both tests resulted in the same conclusion with a failure to reject the null hypothesis. Therefore, there is no significant difference in the mean across all 4 orientation groups, so the recorded flux is most likely due to the star's own properties instead.

References

- Boyajian, T. S., LaCourse, D. M., Rappaport, S. A., Fabrycky, D., Fischer, D. A., Gandolfi, D., Kennedy, G. M., Korhonen, H., Liu, M. C., Moor, A., et al. (2016). Planet Hunters IX. KIC 8462852 – where's the flux? *Monthly Notices of the Royal Astronomical Society*, 457(4), 3988–4004.
<https://doi.org/10.1093/mnras/stw218>
- Montet, B. T., & Simon, J. D. (2016). KIC 8462852 faded throughout the Kepler mission. *The Astrophysical Journal Letters*, 830(2), Article L39.
<https://doi.org/10.3847/2041-8205/830/2/L39>
- Montet, B. T., Tovar, G., & Foreman-Mackey, D. (2017). Long-term photometric variability in Kepler full-frame images: Magnetic cycles of Sun-like stars. *The Astrophysical Journal*, 851(2), 116.
<https://doi.org/10.3847/1538-4357/aa9e00>